BIG MODULAR / WIRED POWER FLOW

LDO-R1 PROTOTYPE / BENCH QUALIFICATION PENDING

USB PD input

USB-C / FIXED9V

VIN9

GND

Autonomous hardware window and retry

U2 WINDOW / SW1 RETRY

VIN9

VPROT

GND

Independent buck converters

U3 MID / U5 +5V

VPROT

MID

P5

GND

Direct bipolar rails

U4 / INTERNAL +13.531/-13.921V

MID

GND

PRAW

NRAW

Quiet bipolar post-regulators

PRAW

NRAW

GND

P12

N12

Board output and power LED

P12

N12

P5

J7 / D10 P5 LED

GND

GND

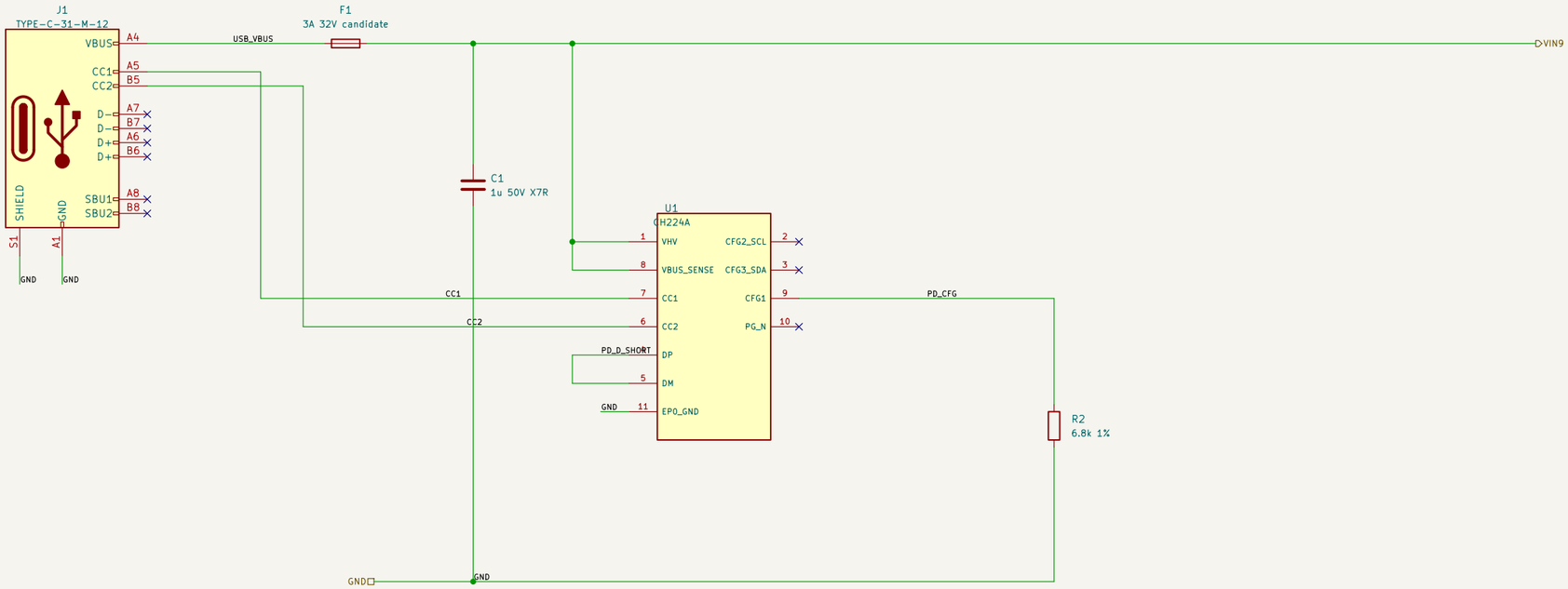
9 V voltage acceptance is NOT proof of a PD protocol contract.
No MCU, AUX regulator, sensing/debug or artificial RUN inhibit.
SW1 retries eFuse; UV/OV window remains independent.

J7 is the physical output boundary: one manual harness.
Top USB/retry/3mm LED are placed together in the native boards.
LDO-R1: quiet +/-12V, capacitor-bank revision; prototype bench qualification pending.

LDO-R1 PROTOTYPE / BENCH QUALIFICATION PENDING		
Big Modular		
Sheet: /		
File: big_modular.kicad_sch		
Title: Big Modular – V1 electrical hierarchy		
Size: User	Date:	**Rev: LDO-R1**
KiCad E.D.A. 10.0.6		Id: 1/7

01 / USB PD INPUT

USB_VBUS -> F1 -> VIN9 (fused, may be 5 V during negotiation).
CH224A fixed 9 V via 6.8k CFG1; CFG2/3 and PG unused. DP/DM shorted locally.
VHV1 and VBUS8 directly VIN9 per WCH table4-1 / section6.1; EP0 -> footprint11.



6.8k single-resistor configuration requests 9 V.
CFG2/3 internal pullups; intentionally unused (no I2C header).
PG intentionally NC: voltage window, not protocol status, permits power.

#FLG01
PWR_FLAG
GND

#FLG02
PWR_FLAG
USB_VBUS

#FLG03
PWR_FLAG
VIN9

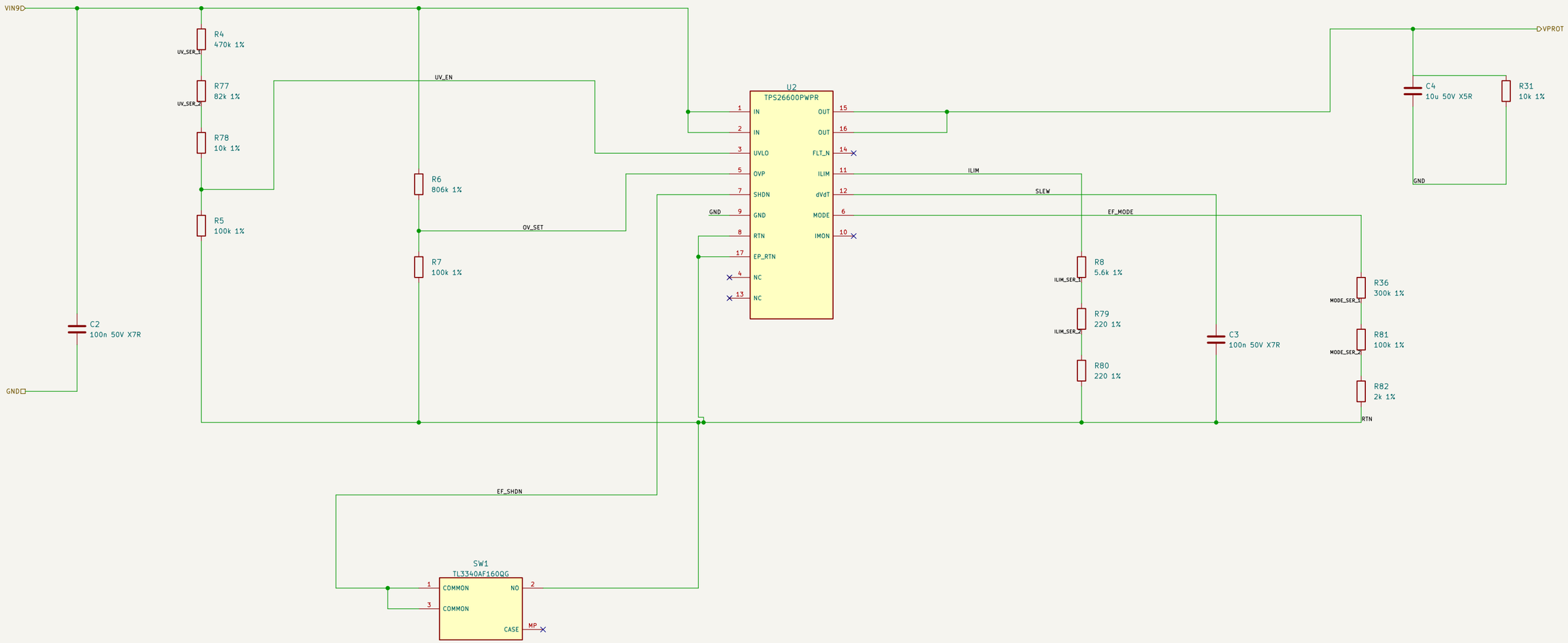
V1 – ROUTED HRO CARRIER / PASSIVE FRONT PANEL
Big Modular
Sheet: /USB PD input/
File: 01_input_protection.kicad_sch

Title: Big Modular – USB PD input

Size: User	Date:	**Rev: V1 routing preparation**
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02 / AUTONOMOUS HARDWARE WINDOW AND RETRY

UV rising 7.878 V / OV rising 10.781 V nominal; hysteresis applies. RTN is NOT GND.
SHDN self-biases internally; SW1 shorts SHDN to RTN while pressed, release resets/retries.
R8 6.04k: 1.987 A nominal. MODE 402k: current limit with thermal latch-off. No PG/MCU permission.



Basic series: R4+R77+R78 = 562k; R8+R79+R80 = 6.04k; R36+R81+R82 = 402k. Each 1%.

SW1 PRESS: SHDN -> RTN, eFuse OFF; RELEASE: internal 2.0..3.4 V bias retries.
No AUX/PG/JP1 or external permission. UV/OV/current/thermal protection always applies.
Bucks use their own internal EN pullups, not this weak SHDN bias.

V1 – ROUTED HRO CARRIER / PASSIVE FRONT PANEL
Big Modular
Sheet: /Autonomous hardware window and retry/
File: 02_hardgate.kicad_sch
Title: Big Modular – Autonomous hardware window and retry
Size: User | Date: | **Rev: V1 routing preparation**
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03 / INDEPENDENT BUCK CONVERTERS

U3 MID 5.007547 V: upper DC corner5.318152V with1%/100ppm/K at100K: +0.1V allocation=5.418152V. U4 ceiling 5.5 V.
U5 direct 5.007547 V from VPROT, never from positive boost. 100n BOOT-to-SW each.
EN pins intentionally floating: TI internal pullups enable from protected VPROT; no SHDN loading.

U3: MID / 5.007547 V

VPROT

C35
100n 50V X7R

U3
TPS54302DDCR
VIN
EN
GND
BOOT
SW
FB

MID_BOOT
MID_SW
MID_FB

L1
6.8uH 20% shielded

MID

VPROT
C68
22u 25V X5R
GND

C5
22u 25V X5R
C7
100n 50V X7R

R9
100k 1%
R10
13k 1%
MID_DIV_LO
R84
510 1%

C6
22u 25V X5R
C37
22u 25V X5R
C70
22u 25V X5R
C71
22u 25V X5R

GND
GND

U5: P5 / 5.007547 V

C36
100n 50V X7R

U5
TPS54302DDCR
VIN
EN
GND
BOOT
SW
FB

FIVE_BOOT
FIVE_SW
FIVE_FB

L4
6.8uH 20% shielded

P5

VPROT
C69
22u 25V X5R
GND

C18
22u 25V X5R
C20
100n 50V X7R

R17
100k 1%
R18
13k 1%
FIVE_DIV_LO
R85
510 1%

C19
22u 25V X5R
C38
22u 25V X5R
C72
22u 25V X5R

GND

BOOT caps are switch-node referenced, not grounded.
Kelvin FB ground to IC GND; keep high dI/dt loops off analog return.
6.8uH Sunlord SWPA80655; validate thermal/loop behavior in V1.

#FLG04
PWR_FLAG
MID

#FLG08
PWR_FLAG
P5

V1 – ROUTED HRO CARRIER / PASSIVE FRONT PANEL
Big Modular
Sheet: /Independent buck converters/
File: 03_bucks.kicad_sch
Title: Big Modular – Independent buck converters
Size: User | Date: | **Rev: V1 routing preparation**
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04 / INTERNAL RAW BIPOLAR RAILS

INTERNAL raw +13.537080 / −13.928806V nominal. MID remains 5.007547V.
P: (1.5M+24k)/150k. N: (1.6M+150k)/(150k+2.4k), referred to BI_REF.
R12/R14 remain0.1%25ppm/K; other divider parts1%100ppm/K. 100mV peak allocation, not measured; conditional dropout.

MID
GND
C16
10u 50V X5R
C17
10u 50V X5R
L2
4.7uH 20% shielded
D2
SS34 40V
BOOST_SW
PRAW
R12
1.5M 0.1%
POS_DIV
R75
24k 1%
R13
150k 1%
C10
4.5p 50V C0G
C8
22u 25V X5R
U4
TPS65131RGER
INP
BSW
INN
VIN
ENP
PSP
ENN
PSN
VPOS
FBP
VREF
FBN
VNEG
OUTN
CN
CP
AGND
EP
PGND
POS_FB
ANA_MID
COMP_N
R16
100 1%
R38
100k 1%
BI_EN
C15
100n 50V X7R
R40
100k 1%
D3
SS34 40V
INV_SW
NRAW
L3
4.7uH 20% shielded
COMP_P
R14
1.6M 0.1%
NEG_DIV
C11
4.3p 50V C0G
C9
22u 25V X5R
R60
150k 1%
C14
4.7n 50V X7R
C13
10n 50V X7R
NEG_FB
R15
150k 1%
NEG_REF_DIV
R76
2.4k 1%
BI_REF
C12
220n 25V X7R

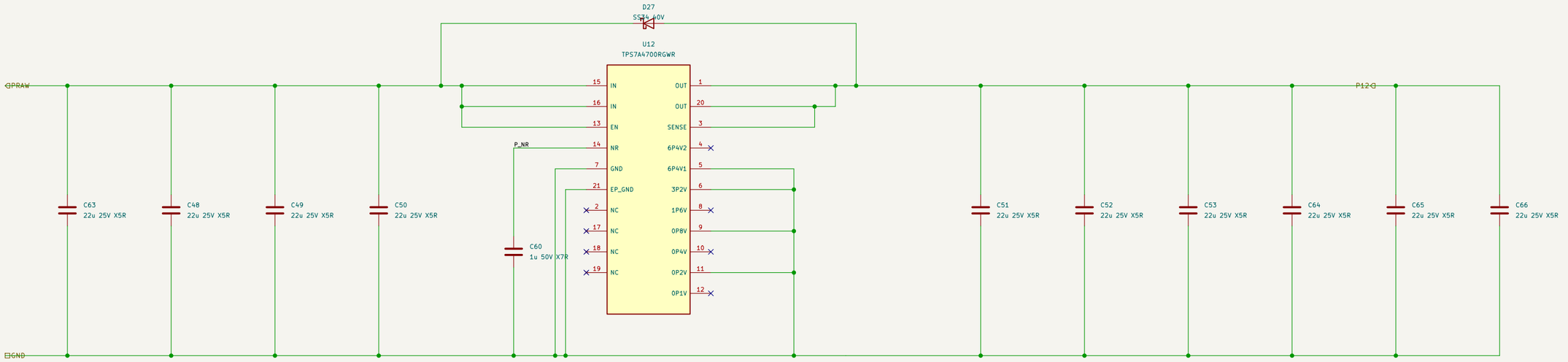
Separate INP/INN local ceramic loops; shielded L2/L3.
No arbitrary output LC. Compensation and Cff require loop validation.

#FLG05
PWR_FLAG
ANA_MID
#FLG06
PWR_FLAG
PRAW
#FLG07
PWR_FLAG
NRAW

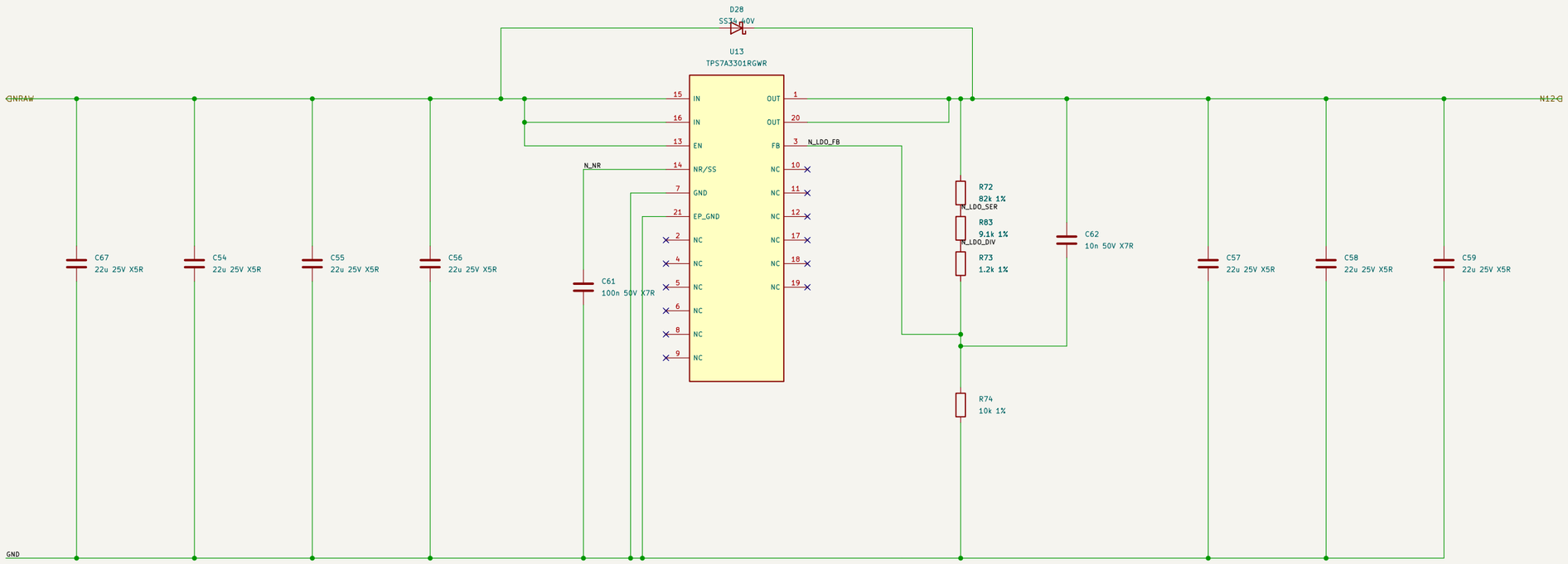
LDO−R1 PROTOTYPE / BENCH QUALIFICATION PENDING
Big Modular
Sheet: /Direct bipolar rails/
File: 04_bipolar.kicad_sch
Title: Big Modular — Direct bipolar rails
Size: User | Date: | **Rev: LDO−R1**
KiCad E.D.A. 10.0.6 | Id: 5/7

05 / QUIET BIPOLAR POST-REGULATORS — LDO-R1 PROTOTYPE

Raw +13.537080/−13.928806V remain internal. Final +5V is unchanged and bypasses this sheet.
Both exposed pads 21 are GND. Exact requested RGWR parts; NOT a +/−1A supply.



U12 ANY-OUT: 5,6,9,11 to GND; 4,8,10,12 float. SENSE3 -> final output.
C60 sees approximately +12V. 4/6/4/3 x22u on IN+/OUT+/IN-/OUT-: typical bias + tolerance/temp/aging screen passes.
Require >=10u effective at each IN/OUT; >=20u at positive OUT for accuracy conditions.



(R72+R83+R73)/R74 =92.3k/10k -> −12.02025V before IFB; 117.5uA nominal divider current (>5uA required).
C61 100n: typical ~120ms start, not rail sequencing. C62 10n improves AC; shutdown FB kick remains a test gate.
D27/D28 divert reverse current, do NOT block it. SS34 Vf does NOT prove the negative −0.3V differential limit.
800mA aspiration is load-dependent, not a simultaneous rail guarantee. Capacitor bias/ESR, transient and board thermal qualification pending.
Authorized Basic1% divider substitutions: full-temperature negative dropout/accuracy is conditional, not a1A guarantee.

LDO-R1 PROTOTYPE / BENCH QUALIFICATION PENDING

Big Modular

Sheet: /Quiet bipolar post-regulators/
File: 05_quiet_rails.kicad_sch

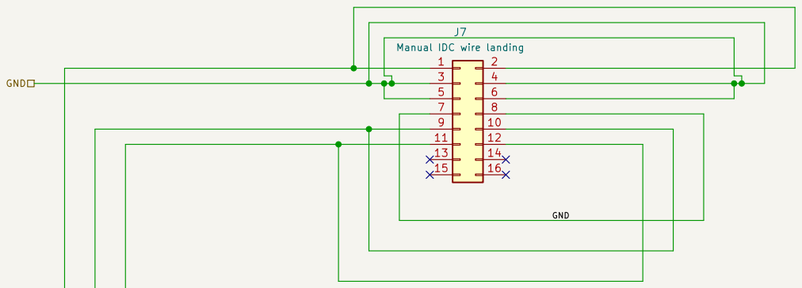
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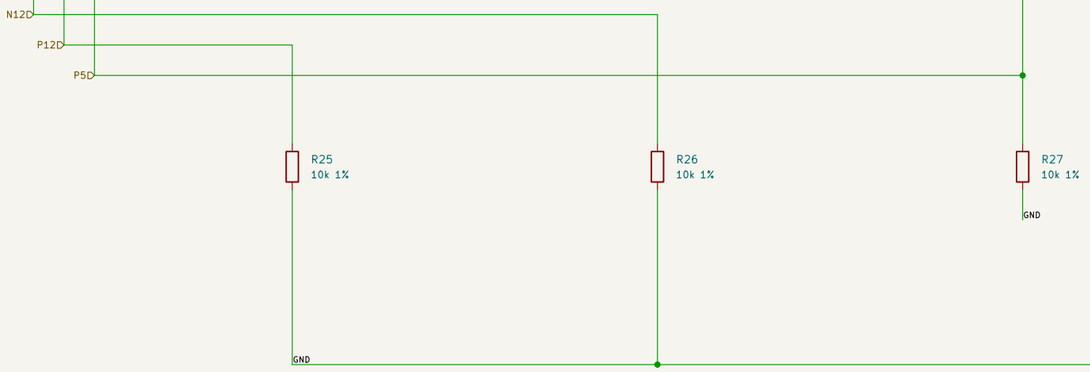
Size: User	Date:	**Rev: LDO-R1**
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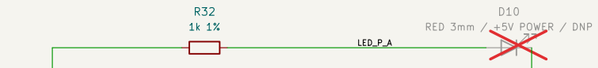
05 / BOARD OUTPUT AND POWER LED

J7 physical board wire landing: 1/2=−12, 3..8=GND, 9/10=+12, 11/12=+5; 13..16 NC.
One manual output harness, no J2 symbol or second source. Never parallel another active PSU.
Single red D10: final P5 -> 1k -> anode2; cathode1 -> GND. +5 V presence only.

J7: PHYSICAL BOARD OUTPUT LANDING







Single red D10: FINAL P5 presence only.
R32 1k: no blink, MCU or all-rail status claim.
Manual harness from J7: 13..16 NC. Never parallel a PSU.

V1 — ROUTED HRO CARRIER / PASSIVE FRONT PANEL
Big Modular
Sheet: /Board output and power LED/
File: 07_output_rails.kicad_sch
Title: Big Modular — Board output and power LED
Size: User | Date: | **Rev: V1 routing preparation**
KiCad E.D.A. 10.0.6 | Id: 7/7